Sinusoidal Signals in Continuous & Discrete Time

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Continuous-Time Sinusoidal Signal — General Form

General Equation

$$\boxed{x(t) = A\cos(\Omega t + \theta)} \qquad \text{or equivalently} \qquad x(t) = A\sin(\Omega t + \theta)$$

- A **Amplitude**
- Ω **Angular Frequency** (rad/s)
- t **Continuous time** $(-\infty < t < \infty)$
- θ **Phase angle** (radians)
- F **Frequency in Hz**
- T_0 **Fundamental Period** (s)

Key Relations

$$\Omega = 2\pi F, \qquad F = \frac{1}{T_0}$$

$$T_0 = \frac{1}{F} = \frac{2\pi}{\Omega}$$

$$\Omega \in (-\infty, +\infty)$$

Term-by-Term Explanation — Amplitude & Angular Frequency

(1) Amplitude A

- Peak value of the oscillation
- Sets the *vertical scale* of the wave
- $A > 0$ always (sign is absorbed into θ)

Example

$x(t) = 3\cos(\Omega t)$: the signal swings between -3 and $+3$.

(2) Angular Frequency Ω

- How fast the signal oscillates, in **radians per second**
- Higher $\Omega \Rightarrow$ more cycles per second
- $\Omega = 2\pi F$

Example

Mains electricity in Bangladesh: $F = 50\,\text{Hz}$

$\Omega = 2\pi \times 50 \approx 314.16\,\text{rad/s}$

Period $T_0 = 1/50 = 0.02\,\text{s} = 20\,\text{ms}$

Term-by-Term Explanation — Phase & Periodicity

(3) Phase Angle θ

- Shifts the signal *left or right* in time
- $\theta > 0$: signal leads (shifts left)
- $\theta < 0$: signal lags (shifts right)
- $\theta = \pi/2$: cos $\to$ sin

(4) Periodicity (CT)

A CT sinusoid is **always periodic** with period:

$$T_0 = \frac{2\pi}{\Omega} = \frac{1}{F}$$

because $\cos(\Omega(t + T_0) + \theta) = \cos(\Omega t + \theta)$ for any t.

Example

$x(t) = \cos\left(\Omega t + \frac{\pi}{4}\right)$
Peak occurs at $t = -\frac{\pi}{4\Omega}$ (earlier than $t = 0$)

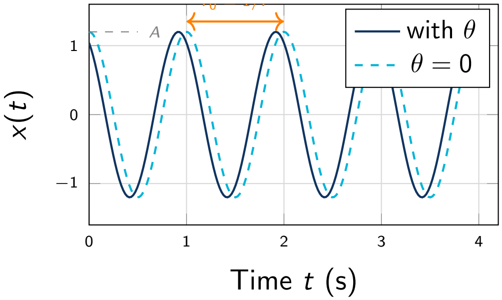
Key Condition

$$x(t + T_0) = x(t) \quad \forall\, t \in \mathbb{R}$$

This holds for **every** CT sinusoid — no exceptions!

CT Sinusoid — Waveform & Properties

$$x(t) = A\cos(\Omega t + \theta)$$



CT Signal Properties

Property	Value
Range	$[-A,\ A]$
Periodic?	**Always YES**
Period	$T_0 = 2\pi/\Omega$
Energy	∞ (periodic)
Power	$A^2/2$
Ω range	$(-\infty, +\infty)$

Discrete-Time Sinusoidal Signal — General Form

General Equations

$$\boxed{x(n) = A\cos(\omega n + \phi)} \qquad \text{or equivalently} \qquad x(n) = A\cos(2\pi f n + \phi)$$

Symbol	Name	Description / Range
n	Sample index	Integer: $n = 0, \pm 1, \pm 2, \pm 3, \ldots \; (-\infty < n < \infty)$
A	Amplitude	Peak value; real positive number
ω	Discrete angular freq.	Radians/sample; $\omega \in [-\pi,\ \pi]$
f	Normalized frequency	Cycles/sample; $f \in \left[-\frac{1}{2},\ \frac{1}{2}\right]$
ϕ	Phase angle	Phase shift in radians
N	Period (if periodic)	Smallest positive integer for $x(n+N) = x(n)$

DT Sinusoid — Sample Index n in Detail

What is n?

The sample index n takes only **integer values**:

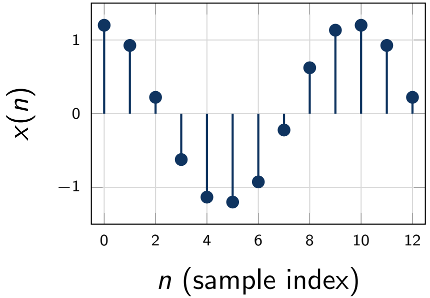
$$n \in \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

Each n represents one **sample** taken at time

$$t = n\,T_s \quad \text{where } T_s = \frac{1}{F_s}$$

F_s is the sampling rate (samples per second).

DT Sinusoid (stems)



Note

Unlike CT, n is NOT continuous. The signal only **exists at discrete instants**.

DT Sinusoid — Sample Index n in Detail

Example

CD-quality audio: $F_s = 44100\,\mathrm{Hz}$

$n = 44100$ corresponds to $t = 1\,\mathrm{second}$

$x(0), x(1), x(2), \ldots$ are the *digital samples*

DT Sinusoid — Normalized Frequency f & ω

Normalized Frequency f

$$f = \frac{F}{F_s}, \qquad f \in \left(-\frac{1}{2}, \frac{1}{2}\right]$$

F = physical frequency (Hz), F_s = sampling rate

DT Sinusoid — Normalized Frequency f & ω

- f is **dimensionless** (cycles per sample)
- $f = 0 \rightarrow$ DC (constant signal)
- $f = \pm\frac{1}{2} \rightarrow$ **highest** representable frequency (Nyquist)
- $f = \frac{1}{4} \rightarrow$ completes one cycle every 4 samples

Discrete Angular Freq. ω

$$\omega = \frac{\Omega}{F_s} = 2\pi f, \quad \omega \in [-\pi,\ \pi]$$

Numerical Example

$F = 1000\,\text{Hz},\ F_s = 8000\,\text{Hz}$

$f = \dfrac{1000}{8000} = \dfrac{1}{8} = 0.125$

$\omega = 2\pi \times 0.125 = \dfrac{\pi}{4}\,\text{rad/sample}$

So: $x(n) = A\cos\left(\dfrac{\pi}{4}n + \phi\right)$

Conditions for Periodicity — Discrete Time

Periodicity Condition for DT Sinusoid

$$x(n+N) = x(n) \quad \Longleftrightarrow \quad \frac{\omega}{2\pi} = \frac{f_0}{N} \in \mathbb{Q} \quad \text{(rational number)}$$

$$\text{i.e., } f = \frac{k}{N} \text{ for some integers } k, N$$

Conditions for Periodicity — Discrete Time

Finding the period N:

Formula for Period

$$\omega N = 2\pi m \;\Rightarrow\; N = \frac{2\pi m}{\omega}$$

N must be the **smallest positive integer** satisfying this for some integer m.

Example 1 — Periodic

$\omega = \pi/6$:

$$N = \frac{2\pi \times 1}{\pi/6} = 12 \text{ samples}$$

$x(n) = \cos(\pi n/6)$ has period $N = 12$

Example 2 — NOT Periodic

$\omega = 1\,\text{rad/sample}$:

$N = 2\pi/1 = 2\pi \notin \mathbb{Z}$

$x(n) = \cos(n)$ is **not periodic!**

Periodicity — Rational vs Irrational f

Is $f = \omega/(2\pi)$ **rational**?

YES

YES
$f = k/N$
DT signal is **Periodic**
Period $= N$ samples

NO

NO
f irrational
DT signal is **Aperiodic**
(Never repeats)

Important contrast with Continuous Time:

	CT Sinusoid	**DT Sinusoid**
Always periodic?	**YES**	**NO — only if f is rational**
Period	$T_0 = 2\pi/\Omega$	$N = 2\pi m/\omega$ (integer)
Frequency range	$\Omega \in (-\infty, \infty)$	$\omega \in (-\pi, \pi]$

Worked Example 1 — Finding Period of DT Signal

Problem

Determine whether $x(n) = 3\cos\left(\frac{3\pi}{7}n + \frac{\pi}{3}\right)$ is periodic. If so, find its period.

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Step-by-step solution:

1. Identify $\omega = \frac{3\pi}{7}$

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Step-by-step solution:

1. Identify $\omega = \frac{3\pi}{7}$
2. Compute $f = \frac{\omega}{2\pi} = \frac{3\pi/7}{2\pi} = \frac{3}{14}$

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3. Check: Is $f = \frac{3}{14}$ rational? **YES** — it is k/N with $k = 3$, $N = 14$

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2. Compute $f = \frac{\omega}{2\pi} = \frac{3\pi/7}{2\pi} = \frac{3}{14}$
3. Check: Is $f = \frac{3}{14}$ rational? **YES** — it is k/N with $k = 3$, $N = 14$
4. Period: $N = \frac{2\pi m}{\omega} = \frac{2\pi \times 1}{3\pi/7} = \frac{14}{3}$ — not integer for $m = 1$

Worked Example 1 — Finding Period of DT Signal

Problem

Determine whether $x(n) = 3\cos\left(\frac{3\pi}{7}n + \frac{\pi}{3}\right)$ is periodic. If so, find its period.

Step-by-step solution:

1. Try $m = 3$: $N = \dfrac{2\pi \times 3}{3\pi/7} = 14$ ✓ (integer!)

Worked Example 1 — Finding Period of DT Signal

Problem

Determine whether $x(n) = 3\cos\left(\frac{3\pi}{7}n + \frac{\pi}{3}\right)$ is periodic. If so, find its period.

Answer

$x(n)$ is **periodic** with fundamental period $N = \mathbf{14}$ samples.

Worked Example 2 — CT Sinusoid from Sampling

Problem

A CT signal $x(t) = 2\cos(2\pi \times 440\, t)$ (A-note) is sampled at $F_s = 8000\,\text{Hz}$. Write the DT signal and verify it is periodic. Find its period.

Worked Example 2 — CT Sinusoid from Sampling

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A CT signal $x(t) = 2\cos(2\pi \times 440\, t)$ (A-note) is sampled at $F_s = 8000\,\text{Hz}$. Write the DT signal and verify it is periodic. Find its period.

Solution:

1. $t = n/F_s = n/8000$, so $x(n) = 2\cos\left(2\pi \times 440 \times \dfrac{n}{8000}\right)$

Worked Example 2 — CT Sinusoid from Sampling

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A CT signal $x(t) = 2\cos(2\pi \times 440\, t)$ (A-note) is sampled at $F_s = 8000\,\text{Hz}$. Write the DT signal and verify it is periodic. Find its period.

Solution:

1. $t = n/F_s = n/8000$, so $x(n) = 2\cos\left(2\pi \times 440 \times \dfrac{n}{8000}\right)$
2. Simplify: $\omega = \dfrac{2\pi \times 440}{8000} = \dfrac{2\pi \times 11}{200} = \dfrac{11\pi}{100}$

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2. Simplify: $\omega = \dfrac{2\pi \times 440}{8000} = \dfrac{2\pi \times 11}{200} = \dfrac{11\pi}{100}$
3. $f = \dfrac{440}{8000} = \dfrac{11}{200}$ (rational) $\Rightarrow$ periodic!

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2. Simplify: $\omega = \dfrac{2\pi \times 440}{8000} = \dfrac{2\pi \times 11}{200} = \dfrac{11\pi}{100}$
3. $f = \dfrac{440}{8000} = \dfrac{11}{200}$ (rational) $\Rightarrow$ periodic!
4. Period: $N = \dfrac{2\pi m}{\omega}$; smallest $N \Rightarrow m$ s.t. $N = \dfrac{200m}{11}$ is integer $\Rightarrow m = 11,\ N = 200$

Worked Example 2 — CT Sinusoid from Sampling

Problem

A CT signal $x(t) = 2\cos(2\pi \times 440\, t)$ (A-note) is sampled at $F_s = 8000\,\text{Hz}$. Write the DT signal and verify it is periodic. Find its period.

Solution:

1. Period: $N = \dfrac{2\pi m}{\omega}$; smallest $N \Rightarrow m$ s.t. $N = \dfrac{200m}{11}$ is integer $\Rightarrow m = 11,\ N = 200$

Answer

$x(n) = 2\cos\left(\dfrac{11\pi}{100} n\right)$, periodic with $N = 200$ samples per cycle.

Summary — All Key Formulae

Continuous Time

$$x(t) = A\cos(\Omega t + \theta)$$

$\Omega = 2\pi F$ $\quad$ $T_0 = 1/F = 2\pi/\Omega$

Always periodic $\quad$ Power $= A^2/2$

ACF $= \frac{A^2}{2}\cos(\Omega\tau)$

Discrete Time

$$x(n) = A\cos(\omega n + \phi) = A\cos(2\pi f n + \phi)$$

$\omega = 2\pi f$ $\quad$ $f = F/F_s$

$\omega \in [-\pi, \pi)$ $\quad$ $f \in [-\frac{1}{2}, \frac{1}{2})$

Periodic iff $f \in \mathbb{Q}$:

$N = 2\pi m/\omega$ (smallest integer)

Critical Differences

CT: $\Omega \in \mathbb{R}$, always periodic

DT: $\omega \in [-\pi, \pi)$, periodic only if $f \in \mathbb{Q}$

DT: ω and $\omega + 2\pi$ give **same** signal

Thank You